

Recall the functions $cov(l)$ and $H(l)$ are defined in (4.3 and 4.4) respectively. For the sake of subsequent elaboration, we rewrite the functions cov and H as

$$(5.3) \quad cov^h(l^h) = cov(l^*, l^h) = cov(l) = \sum_{\sigma \in \mathcal{T}_{3,1}} \widetilde{cov}_\sigma(l_\sigma^h) + \sum_{\sigma \in \mathcal{T}_{4,0}} cov_\sigma(l_\sigma)$$

and

$$(5.4) \quad H^h(l^h) = H(l^*, l^h) = H(l) = cov(l) - 2\pi \sum_{i \in E} l_i,$$

where $\mathcal{T}_{3,1}$ ($\mathcal{T}_{4,0}$, resp.) represents the set of 3-1 type (4-0 type, resp.) tetrahedra in $\mathcal{T}$.

Lemma 5.3. *For $(M, \mathcal{T})$, let $l^h \in \mathcal{L}^h(M, \mathcal{T})$ be a hyper-ideal edge length vector, then*

$$\frac{\partial H^h}{\partial l_i} = -K_i, \quad e_i \in E^h.$$

Proof. For any $e_i \in E^h$, if the hyper-ideal edge e_i is not a side of any 3-1 type tetrahedron in $\mathcal{T}$, then (4-1) in [9] had proved the conclusion. On the contrary, e_i is a side of some 3-1 type hyperbolic tetrahedra, take such a 3-1 tetrahedron (σ, l_σ) , with

$$l_\sigma = (l_{12}(l_{23}, l_{24}, l_{34}), l_{13}(l_{23}, l_{24}, l_{34}), l_{14}(l_{23}, l_{24}, l_{34}), l_{23}, l_{24}, l_{34}),$$

where $(l_{23}, l_{24}, l_{34}) = l_\sigma^h \in \mathbb{R}^3$, then

$$(5.5) \quad \begin{aligned} \frac{\partial \widetilde{cov}_\sigma(l_\sigma^h)}{\partial l_{23}} &= \frac{\partial cov(l_\sigma)}{\partial l_{23}} \\ &= \frac{\partial cov}{\partial l_{23}} + \frac{\partial cov}{\partial l_{12}} \frac{\partial l_{12}}{\partial l_{23}} + \frac{\partial cov}{\partial l_{13}} \frac{\partial l_{13}}{\partial l_{23}} + \frac{\partial cov}{\partial l_{14}} \frac{\partial l_{14}}{\partial l_{23}} \\ &= \alpha_{23} + \alpha_{12} \frac{\partial l_{12}}{\partial l_{23}} + \alpha_{13} \frac{\partial l_{13}}{\partial l_{23}} + \alpha_{14} \frac{\partial l_{14}}{\partial l_{23}} \\ &= \alpha_{23} + \frac{\pi}{3} \frac{\partial}{\partial l_{23}} (l_{12} + l_{13} + l_{14}). \end{aligned}$$

Similarly, we also have

$$(5.6) \quad \frac{\partial \widetilde{cov}_\sigma(l_\sigma^h)}{\partial l_{24}} = \alpha_{24} + \frac{\pi}{3} \frac{\partial}{\partial l_{24}} (l_{12} + l_{13} + l_{14}).$$

and

$$(5.7) \quad \frac{\partial \widetilde{cov}_\sigma(l_\sigma^h)}{\partial l_{34}} = \alpha_{34} + \frac{\pi}{3} \frac{\partial}{\partial l_{34}} (l_{12} + l_{13} + l_{14}).$$

Substitute (5.5), (5.6) and (5.7) into the following calculation of $\partial H^h / \partial l_i$, we have